

Wind, waves and current — the physics behind the hazard map

Companion notes for the HRDPS wind map with San Juan tidal-current overlay (`hrdps_map.html` , build 2026-07-01a → <https://dsleaf1.github.io/hrdps-wind-map/>)

This document explains, with equations and sources, the wave-steepness physics the app uses to flag **wind-against-current** danger, and then documents exactly which NOAA and ECCO datasets feed the wind/current/wave picture — including how the map found the station coordinates on its own.

Part I — The physics

1. Why steepness, not height

What capsizes a kayak is not a wave's height but its **steepness** — the ratio of height to wavelength,

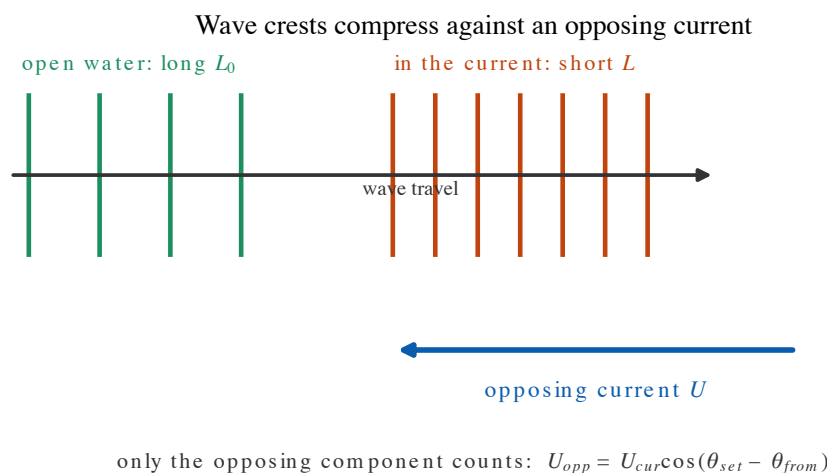
$$S = \frac{H}{L}.$$

A 1-metre swell 100 m long ($S = 1/100$) is a gentle roll; a 1-metre wave 7 m long ($S = 1/7$) is a breaking wall carrying the same height. Waves become unstable and break when the steepness reaches the **limiting steepness** first found by Michell (1893) and Stokes for the highest progressive gravity wave:

$$S_{\max} = \frac{H}{L} \approx \frac{1}{7} = 0.143,$$

at which the crest sharpens to an included angle of 120° and spills. In a real, irregular sea a *significant* steepness of about **1/10** already means a large fraction of the individual waves are breaking — which is why the colour code below treats 1/10 as "very steep" and 1/7 as the outright breaking limit.

The reason opposing current is dangerous is that it drives S upward on **both** counts at once: it shortens L and raises H .



2. Waves on a current: the dispersion relation

Deep-water gravity waves obey the intrinsic (relative-to-the-water) dispersion relation

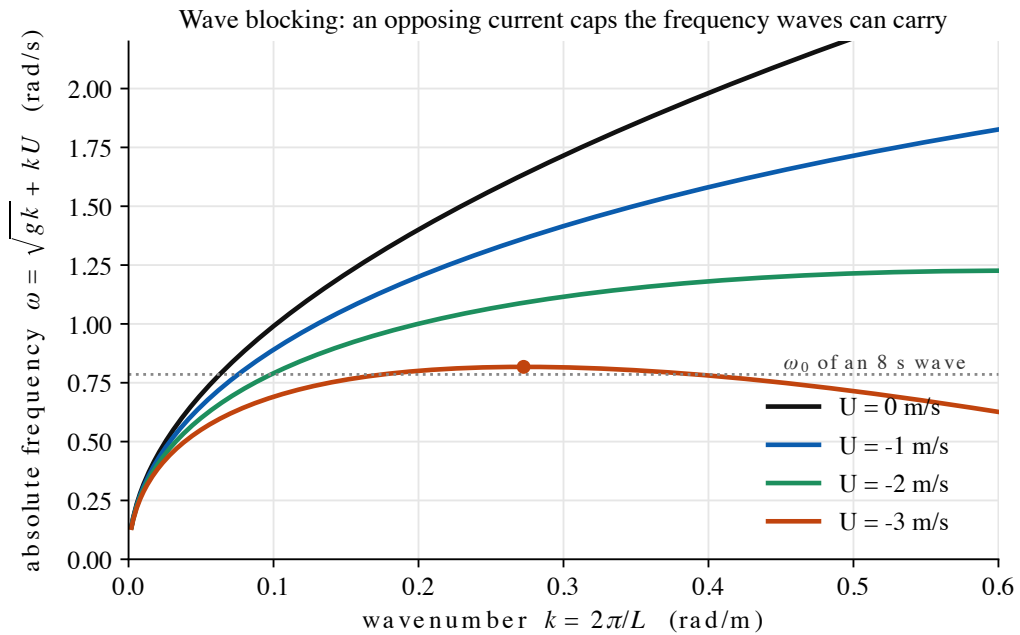
$$\sigma^2 = gk, \quad k = \frac{2\pi}{L},$$

where σ is the angular frequency *seen by an observer drifting with the water* and k the wavenumber. An observer fixed to the seabed instead sees the **absolute** frequency Doppler-shifted by the current U (the component of the current along the wave's direction of travel):

$$\omega = \sigma + kU = \sqrt{gk} + kU \quad (1)$$

For a **steady** current the absolute frequency ω is conserved as the wave train propagates from open water into the current (Longuet-Higgins & Stewart 1961; Phillips 1977). This single conservation law is what forces the wavelength to change.

Equation (1) also contains the whole story of **wave blocking**. Plotting $\omega(k)$ for an opposing current ($U < 0$) shows that the curve stops rising monotonically and develops a maximum: there is a ceiling on the frequency the current can carry.



3. Wavelength shortening and height growth

Let subscript 0 denote the open-water (still-water) values, and define the non-dimensional current and a convenient variable x :

$$\alpha = \frac{U}{c_0}, \quad c_0 = \frac{\sigma_0}{k_0} = \sqrt{\frac{g}{k_0}} = \frac{gT}{2\pi}, \quad x = \sqrt{\frac{k}{k_0}}.$$

Here $c_0 \approx 1.56 T$ (m/s) is the wave's deep-water phase speed. Conserving ω between still water and the current, $\sqrt{gk} + kU = \sqrt{gk_0}$, and dividing through by $\sqrt{gk_0}$ gives a tidy quadratic:

$$\alpha x^2 + x - 1 = 0 \quad \Rightarrow \quad x = \frac{-1 + \sqrt{1 + 4\alpha}}{2\alpha}. \quad (2)$$

Because $L/L_0 = k_0/k = 1/x^2$, an opposing current ($\alpha < 0$, so $x > 1$) **shortens the wavelength**.

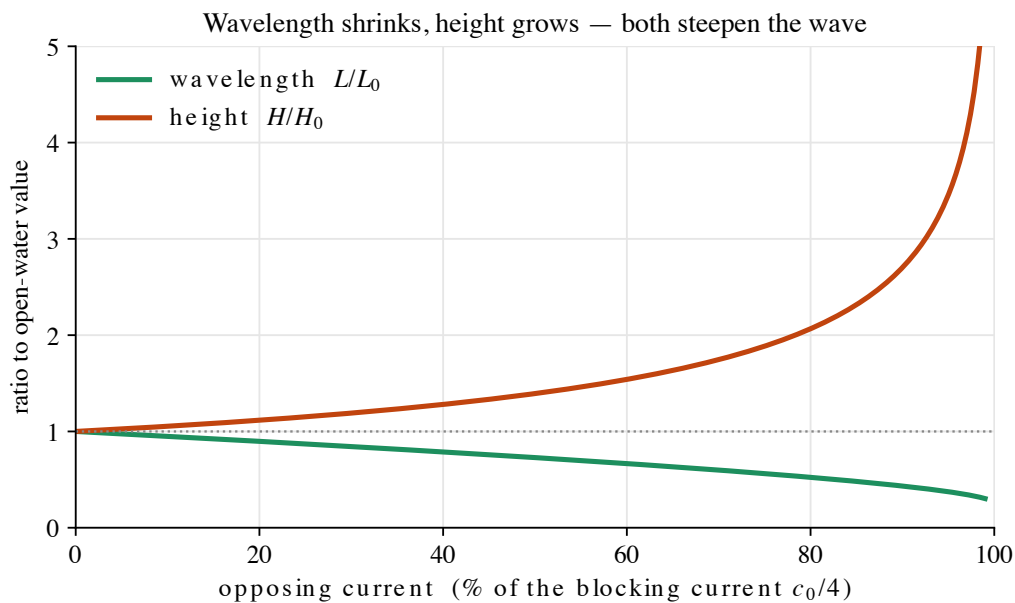
The height follows from conservation of **wave-action flux** (Bretherton & Garrett 1968; Phillips 1977). For steady conditions,

$$(c_g + U) \frac{E}{\sigma} = \text{const}, \quad E = \frac{1}{8} \rho g H^2, \quad c_g = \frac{1}{2} c \text{ (deep water)}. \quad (3)$$

Substituting $\sigma = x \sigma_0$ and $c_g + U = c_0 (2 - x)/(2x^2)$ into (3) yields

$$\frac{E}{E_0} = \frac{x^3}{2 - x}, \quad \Rightarrow \quad \boxed{\frac{H}{H_0} = \frac{x^{3/2}}{\sqrt{2 - x}}}. \quad (4)$$

As the current opposes the waves, the wavelength contracts toward $L_0/4$ and the height grows without bound in linear theory.

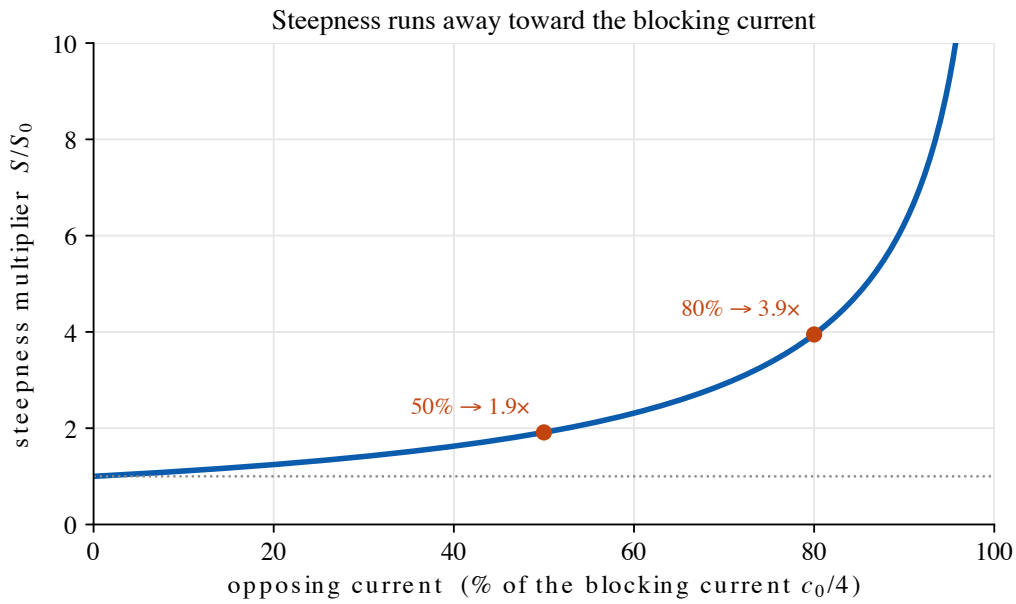


4. The steepness-amplification law

Combining the wavelength change ($L_0/L = x^2$) with the height change (4):

$$\boxed{\frac{S}{S_0} = \frac{H/H_0}{L/L_0} = \frac{x^{3/2}}{\sqrt{2 - x}} \cdot x^2 = \frac{x^{7/2}}{\sqrt{2 - x}}} \quad (5)$$

This is the exact function the app evaluates (`hazBand()` in `hrdps_map.html`). Steepness climbs faster than height because the wavelength is shrinking at the same time. The practical message is the **cliff**: near the blocking current a small increase in current tips the sea over the edge.



opposing current (of blocking)	steepness multiplier S/S_0
20 %	$\approx 1.25\times$
50 %	$\approx 1.9\times$
80 %	$\approx 4.0\times$
95 %	$\approx 9\times$
100 %	$\rightarrow \infty$ (breaks)

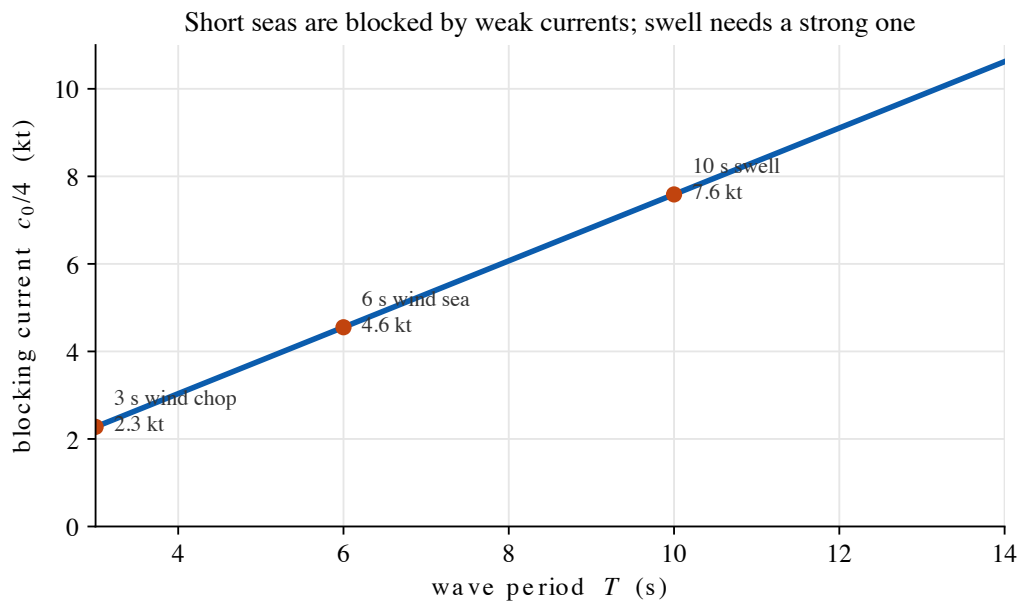
5. Wave blocking at a quarter of the wave speed

Equation (2) has a real solution only while $1 + 4\alpha \geq 0$, i.e.

$$U_{\text{stop}} = \frac{c_0}{4} = \frac{gT}{8\pi} \approx 0.76 T \text{ (kt)} \quad (6)$$

At exactly this opposing current $x = 2$ (so $k = 4k_0$, $L = L_0/4$) and the ground-frame group velocity $c_g + U = 0$: the wave energy can no longer advance, so it accumulates and breaks in place. **This is the mechanism of a tide race or overfalls** (Peregrine 1976; Chawla & Kirby 2002).

The blocking current scales with period, which produces the single most important piece of local knowledge for a paddler:



- A short **3-second wind chop** is blocked by only ≈ 2.3 kt of opposing current.
- A **10-second swell** needs ≈ 6 kt before it blocks.

So the everyday hazard is local wind-chop meeting an ebb; long swell against current produces the dramatic *standing waves*, but only where the flow really rips (Deception Pass, Cattle Point on a big ebb).

6. Only the opposing component counts

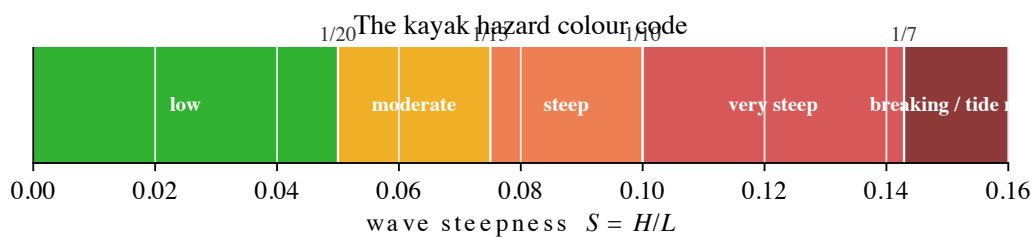
The current speed that matters in (1)–(6) is the component of the tidal set along the wave's line of travel. With the wave *coming from* θ_{from} and the current *setting toward* θ_{set} ,

$$U_{\text{opp}} = U_{\text{cur}} \cos(\theta_{\text{set}} - \theta_{\text{from}}). \quad (7)$$

$U_{\text{opp}} > 0$ opposes the waves (dangerous); $U_{\text{opp}} < 0$ is a *following* current that stretches the waves longer and lower — it actually flattens the sea. A current quartering the waves projects onto their axis and is correspondingly less severe.

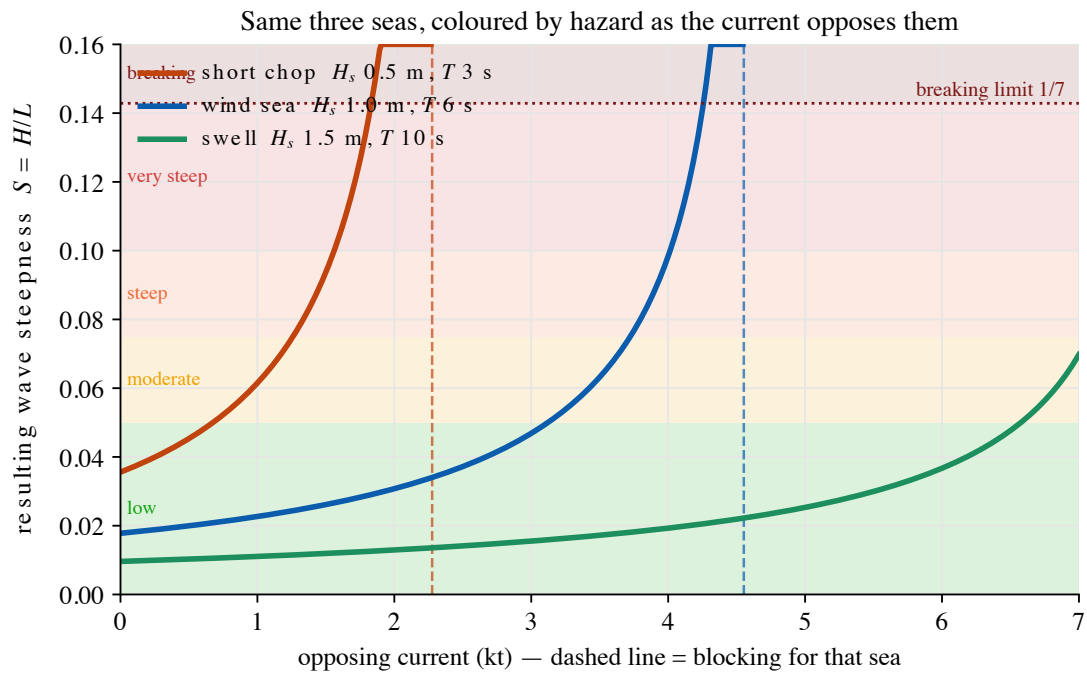
7. From physics to a colour code

The app evaluates (2), (5) and (7) at every current station on the 48-hour timeline, using the live wave state (H_s , T , direction) and the tidal current at that instant, then colours a ring by the resulting steepness S :



Band	Steepness S	Meaning for a kayak
Low	$< 1/20$	Gentle; no breaking
Moderate	$1/20\text{--}1/13$	Whitecaps building
Steep	$1/13\text{--}1/10$	Frequent breaking crests — hazardous
Very steep	$1/10\text{--}1/7$	Widespread breaking
Breaking	$> 1/7$ or blocked	Tide race / overfalls — avoid

Plotting the **resulting** steepness for three representative seas as the opposing current grows shows the colour code in action, and why the short chop is the first to go red:



Consequence gate. Linear theory (4)–(5) diverges at blocking, but a real wave stops growing once it breaks and sheds energy, and a *tiny* sea — a 0.1 m ripple — is low-consequence even when technically "blocked". The app therefore caps the height amplification at 2.5× and holds back the top colours unless the amplified wave height is meaningful:

$$\hat{H} = H_s \cdot \min\left(\frac{x^{3/2}}{\sqrt{2-x}}, 2.5\right), \quad \hat{H} < 0.3 \text{ m} \Rightarrow \text{cap at Moderate}, \quad \hat{H} < 0.5 \text{ m} \Rightarrow \text{cap at Steep}.$$

This stops small blocked ripples from false-alarming red while still lighting up the genuine wind-against-tide chop.

8. Assumptions and limitations

- **Deep water.** Equations (1)–(6) assume $L \ll \text{depth}$. In the passes this is reasonable for the short seas that dominate; it ignores the extra shoaling steepening that also happens over a shallow sill (which would make real overfalls *worse*, not better — so the estimate is conservative there).
- **Steady, uniform current.** The theory is a snapshot; it does not resolve the horizontal shear and eddies of a real race.

- **Linear / monochromatic.** A single H_s , T and direction stands in for the whole spectrum; the consequence gate is a pragmatic correction for the divergence.
- **Wave data resolution.** The global marine model barely resolves the inland passes (§13), so rings are muted deep in the archipelago and most trustworthy at exposed spots and when seas build.

Sources (physics)

1. J. H. Michell (1893), *On the highest waves in water*, Phil. Mag. 36 — limiting steepness 1/7.
2. M. S. Longuet-Higgins & R. W. Stewart (1960, 1961), *Changes in the form of short gravity waves on long waves and tidal currents*, J. Fluid Mech. 8 & 10 — radiation stress and wave–current interaction.
3. F. P. Bretherton & C. J. R. Garrett (1968), *Wavetrains in inhomogeneous moving media*, Proc. R. Soc. A 302 — conservation of wave action.
4. O. M. Phillips (1977), *The Dynamics of the Upper Ocean*, 2nd ed., Cambridge — wave action and the current-modified energy balance.
5. D. H. Peregrine (1976), *Interaction of water waves and currents*, Adv. Appl. Mech. 16 — wave blocking, the $c_0/4$ stopping current.
6. I. G. Jonsson (1990), *Wave–current interactions*, in *The Sea*, vol. 9, Wiley.
7. A. Chawla & J. T. Kirby (2002), *Monochromatic and random wave breaking at blocking points*, J. Geophys. Res. 107(C7) — laboratory confirmation.
8. L. H. Holthuijsen (2007), *Waves in Oceanic and Coastal Waters*, Cambridge — textbook treatment of dispersion on currents and blocking.

Part II — The data

The map layers three independent forecast sources on one 48-hour timeline: **wind** from Environment and Climate Change Canada (ECCC), **tidal currents** from the U.S. NOAA, and **waves** from Open-Meteo's marine model.

9. Wind — ECCC HRDPS

Primary layer (true 2.5 km). ECCC's High Resolution Deterministic Prediction System, *Continental* domain, distributed as GRIB2 on the MSC Datamart:

```
https://dd.weather.gc.ca/today/model_hrdps/continental/2.5km/{HH}/{hhh}/
{YYYYMMDD}T{HH}Z_MSC_HRDPS_WIND_AGL-10m_RLatLon0.0225_PT{hhh}H.grib2
..._WDIR_AGL-10m... (wind direction, degrees true)
```

- **WIND** = 10 m wind speed (m/s → ×3.6 km/h); **WDIR** = direction (° true).
- Native grid: a **rotated** latitude/longitude grid at $0.0225^\circ \approx 2.5$ km (RLatLon0.0225). Four runs per day.

A small pipeline (`hrdps-ingest`, GitHub Actions, 4×/day) decodes the GRIB2 with `cfgrib/ecCodes`, regrids to each map region with a `scipy` k-d-tree nearest-neighbour, byte-packs speed and direction, gzips, and uploads to Cloudflare R2. The browser reads a `manifest.json` plus one `{region}.bin` per region and holds all 48 hours in memory (instant zoom and scrub).

Fallback layer (~3.7 km). If R2 is unreachable the app fetches the same field live from ECCC's GeoMet WCS service as GeoTIFF and decodes it in-browser with `geotiff.js`:

```
https://geo.weather.gc.ca/geomet?SERVICE=WCS&VERSION=2.0.1&REQUEST=GetCoverage
&COVERAGEID=HRDPS.CONTINENTAL_WSPD (and ..._WD)
&FORMAT=image/tiff&SUBSETTINGCRS=EPSG:4326&SUBSET=x(...)&SUBSET=y(...)
&TIME={iso}&DIM_REFERENCE_TIME={run}
```

Inshore option (1 km). The experimental HRDPS *West* 1 km nest is also wired in (GRIB2 only, on the alpha server, runs 00/12 Z):

```
https://dd.alpha.weather.gc.ca/model_hrdps/west/1km/grib2/{run}/{fff}/
CMC_hrdps_west_WIND_TGL_10_rotated_latlon0.009x0.009_{date}T{run}Z_P{fff}-00.grib2
```

Note that the HRDPS grid's own latitude/longitude are read straight from the GRIB2 message metadata — the grid is **self-describing**, so no coordinates had to be supplied by hand for the wind field.

10. Currents — NOAA CO-OPS tidal-current predictions

Predicted currents come from NOAA's Center for Operational Oceanographic Products and Services (CO-OPS) API:

```
https://api.tidesandcurrents.noaa.gov/api/prod/datagetter?product=currents_predictions
&station={ID}&begin_date={YYYYMMDD}&range=96&interval=MAX_SLACK
&units=english&time_zone=gmt&format=json
```

MAX_SLACK returns the sequence of slack-water and peak flood/ebb events. Each record carries everything the overlay needs:

field	meaning	used for
Time	UTC time of the event	timeline placement
Type	slack / flood / ebb	sign of the flow
Velocity_Major	signed speed (kt): + flood, – ebb	current strength
meanFloodDir	mean flood set (° true)	arrow + θ_{set}
meanEbbDir	mean ebb set (° true)	arrow + θ_{set}

Real sample, San Juan Channel (PUG1703), 1 Jul 2026:

```
02:15  slack  +0.01 kt  flood 347°  ebb 168°
05:39  ebb    -3.11 kt  flood 347°  ebb 168°
10:25  slack   0.00 kt
15:35  ebb    -4.10 kt
```

Between events the app reconstructs a continuous speed with a sinusoidal fit (slack → peak → slack), and it takes the flood/ebb **directions straight from NOAA** rather than assuming ebb = flood + 180° (real passes are often 15–30° off that).

11. Waves — Open-Meteo Marine

Wave state at each current station comes from Open-Meteo's marine model, fetched in a single multi-point call (comma-separated coordinate lists):

```
https://marine-api.open-meteo.com/v1/marine?latitude={lat1,lat2,...}&longitude={lon1,...}  
&hourly=wave_height,wave_period,wave_direction&forecast_days=4&timezone=GMT
```

- `wave_height` = H_s (m), `wave_period` = T (s), `wave_direction` = mean direction *from* which the waves come (° true).
- Returned hourly and matched to the wind timeline; sampled **at the NOAA station coordinates**, so the wave, current and wind used in (7) all refer to the same point.

12. How the station coordinates were found — without your input

You asked specifically about this. Two independent mechanisms mean no coordinates had to be typed in:

(a) The HRDPS grid is self-describing. Every GRIB2 message contains its own grid definition (the rotated-pole parameters and spacing), so `ecCodes/ cfgrib` returns the full latitude/longitude arrays for the wind field automatically. The only human-chosen numbers on the wind side are the six regional viewing rectangles.

(b) NOAA publishes a machine-readable station directory. The map discovered the current stations from NOAA's metadata API, which lists *every* current-prediction station with its coordinates:

```
https://api.tidesandcurrents.noaa.gov/mdapi/prod/webapi/stations.json?  
type=currentpredictions  
→ 4 430 stations, each { id, name, lat, lng }
```

Filtering that list to the San Juan / Gulf Islands bounding box (48.3–48.75° N, 122.5–123.25° W) returns **156 candidate stations**, from which 14 key passes were curated (Rosario, San Juan Channel, Cattle Point, Deception Pass, etc.). The `lat / lng` in that directory are what position the arrows and rings on the map, and the same coordinates are handed to Open-Meteo for the wave sample. So the whole San Juan overlay is bootstrapped from one public station list plus a bounding box — no manual coordinate entry.

13. The 14 curated San Juan stations

NOAA ID	Pass	Lat	Lon
PUG1702	Rosario Strait	48.458	−122.750
PUG1703	San Juan Channel (S entrance)	48.461	−122.952
PCT1416	Cattle Point (2.8 mi SSW)	48.400	−123.000
PUG1629	Deception Pass (Yokeko Pt)	48.413	−122.613
PUG1734	Guemes Channel (W entrance)	48.521	−122.652
PUG1733	Thatcher Pass	48.527	−122.804
PUG1705	Obstruction Pass	48.603	−122.813
PUG1704	Peavine Pass (W entrance)	48.587	−122.819
PUG1719	Spieden Channel	48.628	−123.112
PUG1715	President Channel	48.673	−123.006
PUG1739	Bellingham Channel (S)	48.540	−122.683
PUG1721	Wasp Passage narrows	48.593	−122.990
PUG1723	Upright Channel narrows	48.554	−122.923
PUG1722	Harney Channel	48.590	−122.922

14. Putting it on one clock

The R2 wind manifest defines the hourly UTC timeline. NOAA current events are placed on it by their UTC timestamps and interpolated; Open-Meteo waves are matched to the nearest hour. Scrubbing or playing the 48-hour slider therefore advances wind, current and the wave-against-current hazard ring together — which is the whole point: you can see the ebb turn against the afternoon sea-breeze chop and watch a pass go amber → red in step.

Sources (data)

- ECCC / MSC — HRDPS product page and Datamart: https://eccc-msc.github.io/open-data/msc-data/nwp_hrdps/readme_hrdps_en/ · <https://dd.weather.gc.ca/>
- ECCC GeoMet (OGC WMS/WCS): https://eccc-msc.github.io/open-data/msc-geomt/readme_en/
- NOAA CO-OPS API (data + metadata): <https://api.tidesandcurrents.noaa.gov/api/prod/> · <https://api.tidesandcurrents.noaa.gov/mdapi/prod/>
- Open-Meteo Marine Weather API: <https://open-meteo.com/en/docs/marine-weather-api>

Figures generated from the same equations the app evaluates (`figs/fig1...7_*.svg`). Symbols: H height, L wavelength, $k = 2\pi/L$, T period, $S = H/L$ steepness, σ intrinsic frequency, ω absolute frequency, U opposing current component, $c_0 = gT/2\pi$ deep-water phase speed, $\alpha = U/c_0$, $x = \sqrt{k/k_0}$.